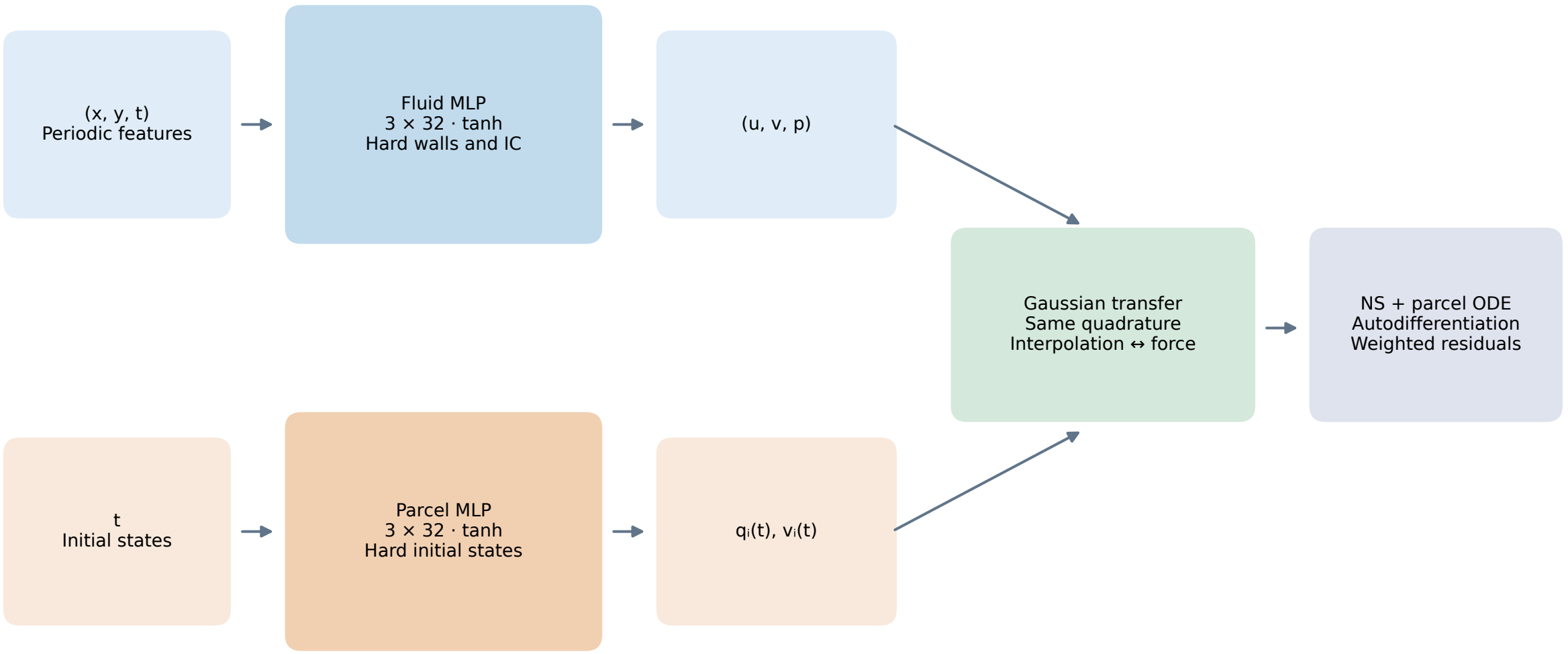
Coupled PINN
(x, y, t)
Periodic features
Fluid MLP
3 × 32 · tanh
Hard walls and IC
(u, v, p)
t
Initial states
Parcel MLP
3 × 32 · tanh
Hard initial states
$q_i(t)$, $v_i(t)$
Gaussian transfer
Same quadrature
Interpolation ↔ force
NS + parcel ODE
Autodifferentiation
Weighted residuals
Physics-only training • held-out reference data • independent residual audit